

Formation of protein self assembled monolayer monitored by Quartz Crystal Micro-balance

Advanced Laboratory Course for Master Student, WS 2022

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0.1 Introduction

Quartz crystal is almost ubiquitous. Quartz Crystal Microbalance (QCM) is an extremely sensitive mass balance that measures in the range of microgram level changes in mass per unit area. In our ongoing lab we are going to find the masses of Cytochrome c, ethanol and do some calibration of quartz crystal itself. In order to find this, we exploit the characteristic frequency of quartz which is mass dependent. If the mass attached with the crystal changes, its frequency will change and then by using Sauerbrey equation we can find the mass of attached substance. Hence the name crystal microbalance.

0.2 Theoretical Background

Piezo-electricity:

It is the generation of electrical-charge on the surface of some crystal by applying some mechanical stress. This is due to the fact that mechanical stress helps in creating microscopic dipoles which adds up to build a considerable macroscopic voltage. Its reverse is also possible – that is to deform certain crystals by applying electric field. This effect is known as reverse piezoelectric effect. Here we are considering small deformation to be able to be in the linear limit. We can write the polarization $\vec{P}$ in the limit of small deformation $\vec{X}$ as

$$\vec{P} = e\vec{X} \quad (1)$$

Where 'e' is the coefficient of piezoelectric polarization. The reverse piezoelectric effect can be written as

$$\vec{X} = d\vec{E} \quad (2)$$

Where d is the coefficient of piezoelectric expansion and $\vec{E}$ corresponds to the applied electric field. The quartz crystals are vapor-deposited with gold electrodes. When the electric field is applied through these electrodes then the crystal develops shear strain.

Resonator

When an electric field is applied to quartz crystals it starts to oscillate with certain eigenfrequency. This creates standing wave within in the crystal. The vibration is called thickness-shear- mode (TSM) as the top and bottom half of the crystal move in opposite directions. The resonance frequency depends on temperature, pressure, and bending stress. In order to minimize the temperature-frequency coupling we use the AT-cut crystals of quartz. AT describes the angle with respect to the crystallographic z-axis on which crystals are cut from the bigger crystal. It helps in stabilizing the frequency at room temperature mainly from $0 - 50^0C$. The voltage as well as the frequency are noted with the electrodes vapor deposited on the crystal. The natural frequency f_0 is given by

$$f_0 = \frac{nK_R}{d_q} \quad (3)$$

Where d_q is the plate size, K_R is the frequency constant of the quartz crystal depending on cut angle of crystal (for AT-cut: $=1664 \text{ m/s}$). Here we are considering the ground state vibration ($n=1$) and its odd harmonics only. The mass sensitivity of the quartz crystal is maximum at the middle and minimum at the border which comes theoretically by using Bessel function but for experiments we use Gauss function.

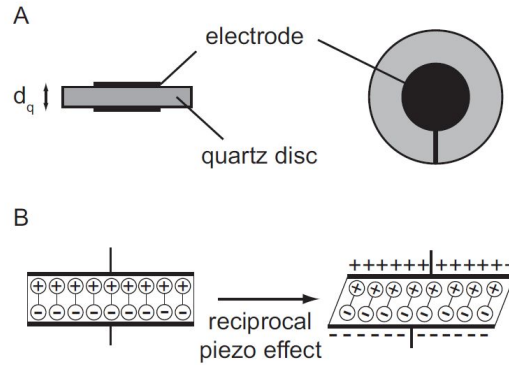


Figure 1: A: a wafer of quartz crystals with side view on the left and top view on the right. B: shear deformation of quartz crystals by application of electric field and the effect of it on dipoles (right) [1]

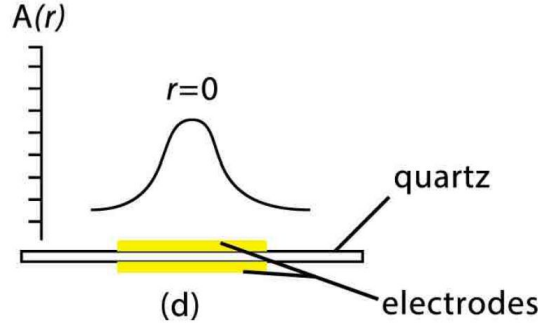


Figure 2: Schematic representation of the variation of amplitude $A(r)$ with the distance (r) from the center of the sensor [?]

Electrical and mechanical description of quartz vibration:

The quartz crystal can be thought same as a damped harmonic oscillator exhibiting the differential equation

$$m \frac{d^2 x}{dt^2} - b \frac{dx}{dt} + kx \quad (4)$$

where 'm' is the mass, 'b' is the damping constant and 'k' is the spring constant.

As we know here, the driving quantity for the quartz crystal is AC voltage, so the final equation analogous to equation (1) is

$$L \frac{d^2 Q}{dt^2} - R \frac{dQ}{dt} + \frac{1}{C} Q \quad (5)$$

where, 'L' defines the inductance and plays the role of mechanical mass of the quartz crystal, 'R' the resistance is same like the damping constant and 'C' the capacitance of given circuit is analogous to spring constant of mechanical system.

Van Dyke and Butterworth gave the representation of mechanical systems driven by electrical field by simple electrical parameters of a circuit and physical properties of piezo-electric resonator see figure-3[1]. In parallel to LCR element is a ' C_0 ' which defines the electrode capacitance of a quartz crystal.

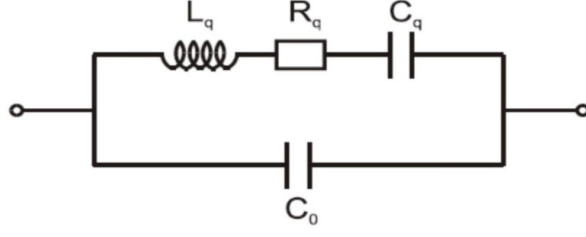


Figure 3: Butterworth-Van-Dyke circuit .jpeg [3]

The oscillation frequency depends upon the mass of a probe, so by depositing the mass the frequency of oscillation changes and it can be described by the Sauerbrey equation

$$\Delta f = -\frac{f_0}{\rho_q \mu_q} \frac{\Delta m}{A} = -S_m \frac{\Delta m}{A} \quad (6)$$

where ρ_q is the density, μ_q the shear modulus, S_m a sensitivity factor and A the area of the quartz crystal. Since the Sauerbrey equation only holds for measurement of quartz in air, vacuum, that resulted not using this technique in liquids for long time. But Kanazawa and Gordon solved the Navier Stokes equation and came up with the idea of standing waves in liquids

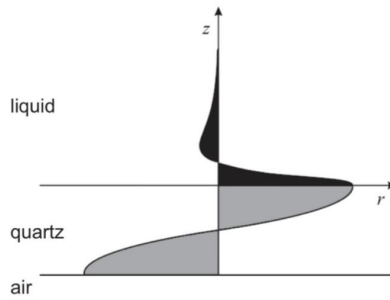


Figure 4: The depth of penetration of wave in water vs air [1]

and these waves are described by the formula

$$\frac{d\nu_r}{dZ^2} = \frac{\rho_L}{\eta_L} \frac{d\nu_r}{dz} \quad (7)$$

where the velocity of the wave is described by ν_r and ρ_L , η_L are the density and the

viscosity of the liquid. As a result, the solution of equation (4) displays that the amplitude of the wave is damped the fairer it spreads in the liquid. The penetration depth of a wave in a liquid is given by

$$\frac{1}{K_L} = \sqrt{\frac{\eta_L}{\pi f_0 \rho_L}} \quad (8)$$

By depositing liquid mass on a quartz crystal the mass is increasing, as a result the oscillation of vibration that is frequency is also changing with it that can be put in equation as

$$\Delta f = -f_0^{\frac{3}{2}} \sqrt{\frac{\eta_L \rho_L}{\pi \mu_q \rho_q}} \quad (9)$$

Cytochrome Sample

A cytochrome C is a protein found in Eukaryotes that includes animals, plants, fungi, protozoans, etc. According to studies the cytochrome of horses is different as compared to humans. It consists a chain of 104 amino-acids, where amino-acids are defined as basic building blocks of all proteins. Cytochrome is present in mitochondria works as an electron carrier At the time of transport of electrons cytochrome c binds to cytochrome c oxidase, such that the electrons are transferred from cytochrome bc_1 complex to cytochrome c oxidase. At the same time, two protons are transmitted from cytochrome bc_1 complex into the inter-membrane space and cytochrome c is oxidised. In contact with a solution ethanol, the surface active substance builds a self-assembled mono-layer. In this experiment, the kinetic energy of binding of cytochrome c is observed, as well as the exhibited frequency change by the QCM measurement[1].

0.3 Experimental Setup

A Teflon holder has been used to host the main component of the QCM experiment. The holder holds the crystal and assist in electrical connection, an oscillator to have the resonance frequency and the digital interfaces that reads as well as store the experimental data.

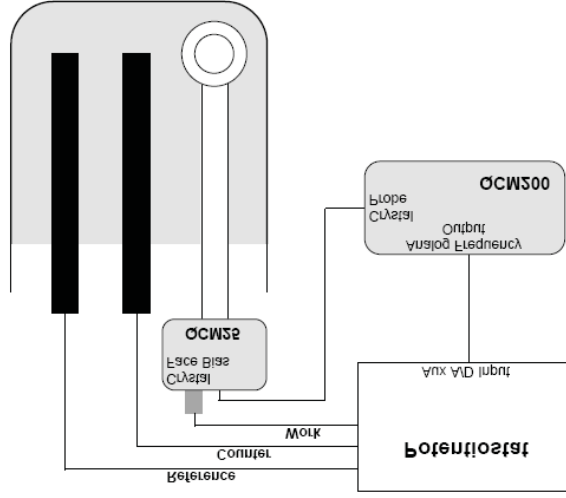


Figure 5: Illustrative diagram of QCM with all necessary equipments [1]

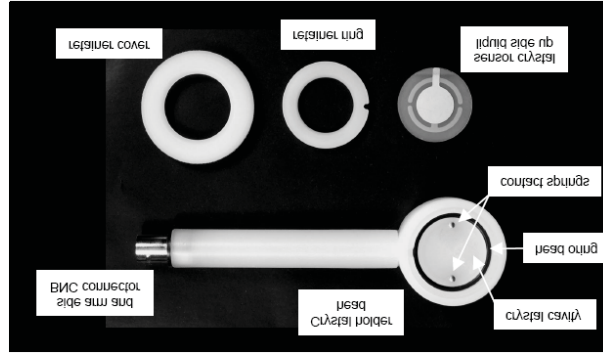


Figure 6: Schematic representation of Teflon holder containing crystal with all essential components [1]

The setup consists of key-hole shaped electrodes deposited in both front and back of the quartz crystal made of gold, a reference electrode made of Ag/AgCl and a counter made of Platin. It's always necessary to ensure the tightness of the crystal and working electrode attachment to avoid unexpected situations. The resonator is usually thicker at the center which helps QCM to oscillate at its center only. As the stability of the frequency is heavily influenced by the changes of temperature, the crystal holder is placed inside a specially designed tempered bath. Several components of the chamber can be adjustable depending on the experiment being done on air/vacuum or liquid [2].

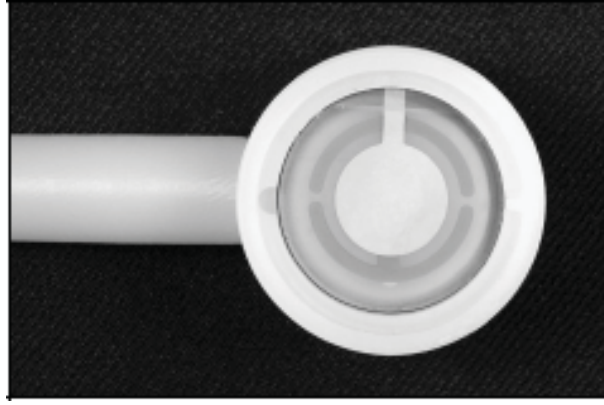


Figure 7: Graphical representation of the quartz within the Teflon holder [1]

0.4 Experimental Procedure

Experimental procedures are as follow.

1. The Experimental procedure begin with cleaning the crystal coated with gold through incubation in a solution called Nonchromix solution about 5 minutes and washing it with miliQ water. Need to pay proper attention at this process since the solution is mix of high concentrated H_2SO_4 .
2. Now after installing the quartz crystal inside the QCM system, the initial task is to determine the eigen-frequency and layer thickness of the crystal.
3. The QCM head is then submerged into tempered waterbath which contain ethanol solution. After certain time, the frequency get stable. Now comparing the experimental and the theoretical data (Kanazawa and Gordon) of the frequency.
4. Since we are interested to examine the layer thickness; 11-Mercaptundecanic acid (MUA) is added into the ethanol (500 μL), so that it can form a MUA layer on the gold surface. The ultimate concentration of the solution must not exceeded 50 μM . Now Measuring the frequency again.

5. The QCM is now put into the buffer solution (10mM KH₂PO₄/K₂HPO₄, pH 7.5) and the cytochrome c solution((500 μ L) is added to observing the binding of C cytochrome provided that frequency is stable.Now measuring the frequency again and determining the exact number of attached molecules by voltammogram.
6. At last, inserting the reference(Ag/AgC) and platin counter electrode beside the gold electrode as well as a potentiostat connecting QCM interface.Taking a total of five cycles with a scan velocity of 50 mV/s in the limit of -.3v to .3v.

0.5 Data analysis

0.5.1 Eigenfrequency and layer thickness of the quartz crystal

The very initial task is to measure the eigenfrequency in air and frequency changes into ethanol and observing the deviation of experimental value from the theoretical value.The eigenfrequency which is in our case is,

$$f_{air} = (5016009.7 \pm 0.005) \quad (10)$$

The measured temperature was 22.4°C that differs a few degree from the room temperature condition.Now having a closer look at the graph depicts that there is a frequency drop at exactly as t= 460s due to submerged the quartz crystal into 100ml ethanol solution and this is due to not having an equilibrium state at instant.The eigenfrequency is nearly constant before the time t=460s and again got stable after t=522s. Since the eigenfrequency is known, we can now calculate the thickness of the quartz by using the following equation.

$$f_{air} = \frac{nK_R}{d_q} \quad (11)$$

Here $K_R = 1664ms^{-1}$, which is the frequency constant of the quartz crystal(AT cut),Since it's associate with grand state vibration, so n= 1.This give a thickness of,

$$d_q = (0.3317 \pm .002)mm \quad (12)$$

The resonance frequency into the ethanol can have by taking the mean of the frequencies in the time range from $t=460s$ to $t=522s$ which in our case is,

$$f_{ethanol} = (5015123.4 \pm 1.003)Hz \quad (13)$$

So the frequency shift between air and ethanol can be found by calculating the difference between the eigenfrequencies of air and ethanol which comes out $-886.3Hz$. Now for the theoretical value can be calculated by using equation (9) that turns out $-709.17Hz$. Here in equation (9); We used $\eta_{Ethanol} = 1.0992mBar.s^2$, $\rho_{Ethanol} = 788.787kgm^{-3}$, $\mu_q = 2.947.10^{13}kgm^{-1}s^{-2}$ and $\rho_q = 2.648 \times 10^3kgm^{-3}$ [4]

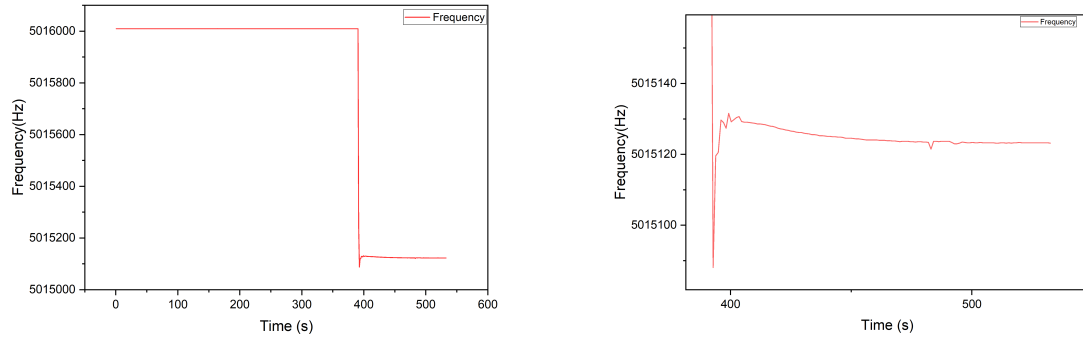


Figure 8: Changes of the resonance frequency while transitioning from air to ethanol(left) and zoom into the range of frequency shift(right)

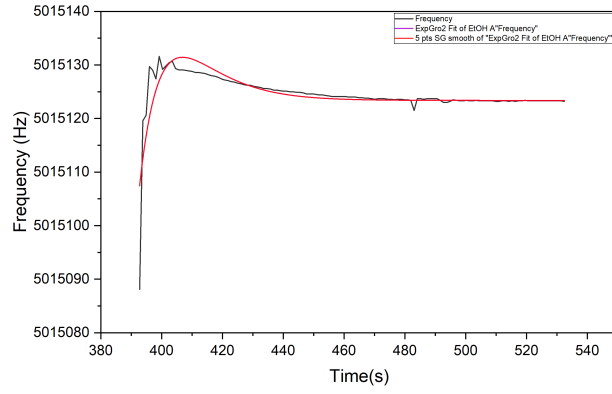


Figure 9: Changes of frequency with exponential fit

0.5.2 Measuring MUA monolayer formation and changes of eigen-frequency

After calculating the frequency difference between eigenfrequency of air and ethanol our next task is to put MUA in ethanol to see its kinetics of binding on quartz crystal. So, we put 1ml of 5mmoler MUA in 100ml of ethanol and got a solution of $50\mu\text{moler}$. In the

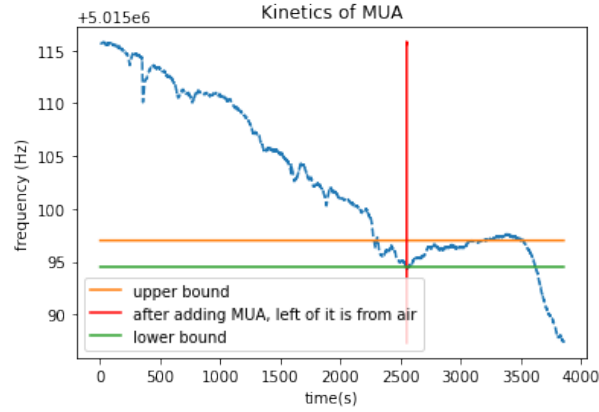


Figure 10: Resonance frequency changes in sample when MUA is added at around 2550s. WE can see the and it examined through upper and lower bound

above figure,we observe that the disturbance created at the time of 2250s is because of adding MUA in beaker containing quartz crystal and ethanol. Then it is really hard to see the point of saturation that means the stabilization of curve with time. We observed

with this idea that if there is slow decrease in frequency in the range of 2600s to 3200s and we took it a range of saturation and calculated the average value for the eigenfrequency of MUA, we found it equal to

$$\Delta f = (3.0 \pm 0.1)Hz$$

$$\implies \Delta m = (66.25 \pm 0.01)ng$$

Here Δm is calculated using equation (6) which shows us the mass deposition that is MUA deposition on quartz crystal. The value calculated may or may not be approximately correct, as we have seen unexpected change of frequency over time. Number of molecules of MUA deposited on quartz crystal = $(1.82 \pm 0.03) * 10^{14}$. That means the number of molecules per unit area is $(1.44 \pm 0.04) * 10^{14}$

0.5.3 Binding of Cytochrome c to the MUA monolayer

The plot of the data we got from adding cytochrome c solution ($500\mu L$) to the buffer solution KH_2PO_4/K_2HPO_4 of concentration 10mM are shown in Fig. 9. After putting the buffer we waited for around 5 minutes so that the curve stabilizes and then after we put the cytochrome c solutions into the beaker. We can see the drift in the curve at time 750s from the plot. Before that we can see a very steep fluctuation at the boundary between the buffer and the cytochrome c addition. This may be due to the disturbance created by adding cytochrome c to the buffer. Since before adding the cytochrome c the curve has not reached equilibrium but there was drift which was not easy to subtract so we moved with the same data.

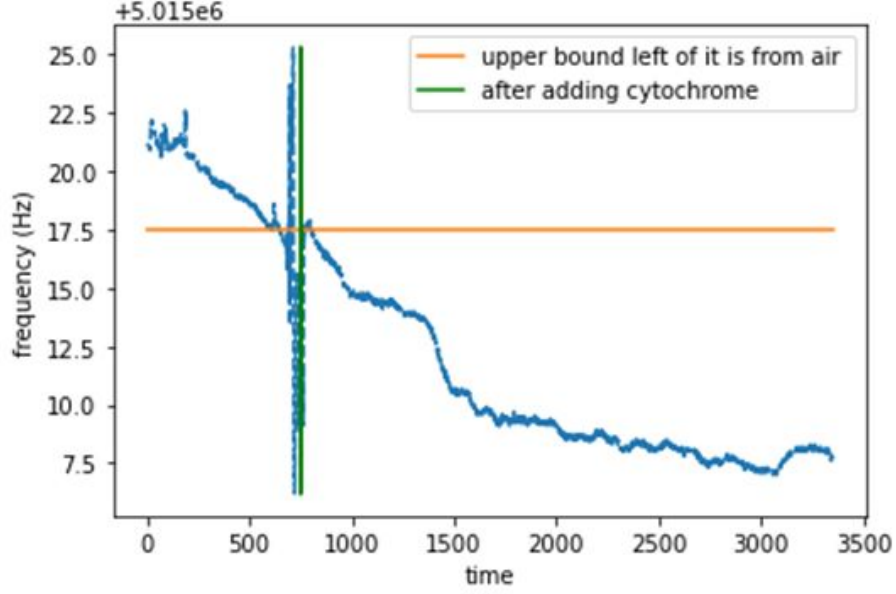


Figure 11: Kinetics of binding of cytochrome. We can observe a drift from linearity. Also we have added Cytochrome c at 750s

After adding cytochrome c initially there was a very sharp drift between frequencies this may be I think due to the change in contact with the electrodes. After that it falls gradually and slowly reaches to the equilibrium. But we did not see the exact horizontal line of reaching equilibrium but it falls very slowly towards the end. We calculated the change in frequency before and after adding cytochrome c after finding two optimum points(upper bound and lower bound) in the plot as shown in figure 10. These optimum points can be considered as resonance of quartz crystal before and after addition of cytochrome.

So, the frequency change we found as:

$$\Delta f = 4.30 \pm 0.50 Hz$$

So after using equation (6) we determined the value of Δm equal to:

$$\Delta m = (95.5 \pm 0.5) ng$$

Now we calculate the number of protein molecules deposited on the surface of quartz crystal. We know the molecular weight of a cytochrome c $M_{cytochrome} = 12,588 DA =$

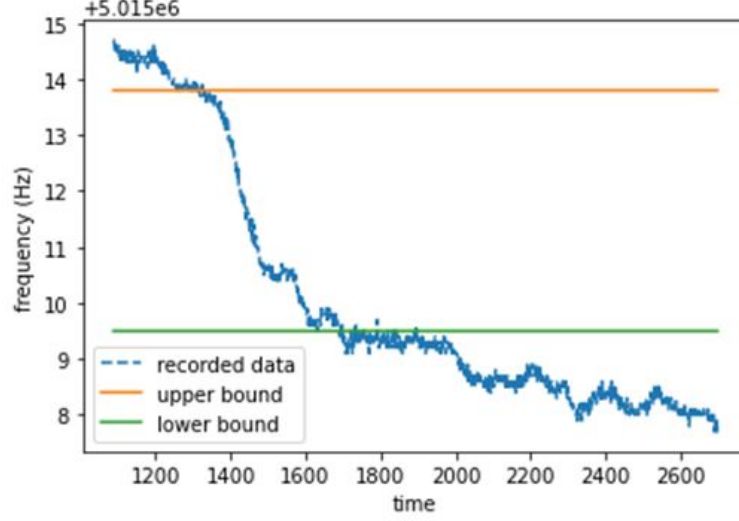


Figure 12: Cytochrome c kinetics. Here we have taken two optimum points to calculate the change in frequency which in the figure is represented by upper and lower bound

12,588g/mol so the expression to calculate number of molecules of cytochrome c deposited on the quartz crystal can be written as:

$$n_{cytochrome} = \frac{\Delta m}{M_{cytochrome}} \implies n_{cytochrome} = \frac{(95.5 \pm 0.5)ng}{12588DA}$$

$$\implies n_{cytochrome} = (4.57 \pm 0.06) * 10^{12}$$

Now we want to see the number of molecules deposited per unit area on the cathode. Therefore we divide the number of molecules by the total area. From this we obtain the value of $(3.61 \pm 0.05) * 10^{12}$ molecules per cm^2 area.

0.5.4 Cyclic Voltammetry measurement of Cytochrome C

The very last part of the experiment is to conduct the cyclic voltammetry for having the proper understanding of the whole kinematics. This process starts with putting two electrodes into the buffer solution and connected with a potentiostat where the working electrode is directly linked with quartz sample.

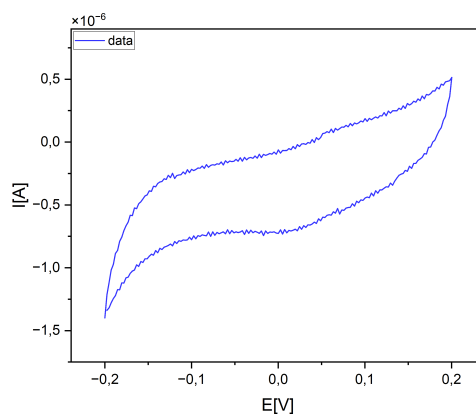


Figure 13: Illustrative.....

0.6 Conclusion

The experimental results are as not as we expected. The measured eigen-frequency changes from air to ethanol which differ 25% from the theoretical value. There are also variation observed while measuring the binding of MUA and Cytochrome to the MUA. Possible factor could be; fluctuation in temperature, probably the electrodes aren't positioned properly with the crystal, the amount of MUA and Cytochrome taken was not as literature. Also the cyclic voltammetry data we worked on was not as expected which showed a large variation while plotting that ultimately causes experimental error.

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